

KAN KUSAKABE

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<https://kankusakabe.github.io/Portfolio>

PROFILE / OBJECTIVE

Ph.D. Candidate in Computer Science with a strong background in Human-Computer Interaction (HCI), software development, and data analysis. Skilled in rapid prototyping, UX research, machine learning, sensing, and video creation. Seeking opportunities in **Software Engineering**, **Data Science**, or **UX Research** roles, leveraging research-driven problem solving and practical development expertise. ([Github](#))

TECHNICAL STRENGTHS

Domain: Rapid Prototyping, 3D Printing, Mobile Application, Machine Learning, VR/AR Development

Languages: Python, C, C#, JavaScript, PHP, Swift, Objective-C, Kotlin, Java, Arduino, Processing, Dart

Frameworks & Libraries: TensorFlow, PyTorch, Next.js, Flutter, React, Unity

Platforms & Tools: Android, iOS, Web (Frontend/Backend/Database), macOS, Arduino, Raspberry Pi, VR

Design & Prototyping: Adobe (Illustrator, Photoshop, Premiere Pro, After Effects, Audition), Figma, Fusion 360

RESEARCH INTERESTS

Human-Computer Interaction (HCI); Interaction Techniques for Mobile Devices; Gesture Design; Sensing Techniques; Machine Learning; 3D Printing; UI/UX Research Methods.

SELECTED EXPERIENCE & PROJECTS

LY Corporation — Research Intern

Jul 2024 – Mar 2025

(Kotlin, Python, PyTorch, 3D Printing)

During this internship, I carried out advanced research on accessory-based smartphone interfaces, extending the concept of RingSense to new form factors. I investigated IMU and touchscreen-based gesture interactions, implemented CNN models to evaluate recognition accuracy, and built a custom Android app for data collection. Additionally, I fabricated prototypes using a 3D printer and organized user workshops, employing open-coding to identify usage scenarios. This end-to-end experience allowed me to combine hardware prototyping, software development, and user-centered research methods.

Personal Finance Web Application

Jan 2025 – Present

(JavaScript, Firebase, Google Authentication)

I created a web application to manage personal and small business finances, initially tailored to my family's needs. By integrating Firebase for secure data management and Google Authentication for login, I enabled straightforward budget tracking and data visualization. Through this self-directed project, I deepened my understanding of both frontend and backend web technologies. ([Link](#))

All Japan Judo Federation — Freelance Multimedia Editor

Dec 2024

(Adobe Premiere, After Effects)

I produced standardized video-editing templates for the federation's podcast channel, enhancing audience engagement and achieving thousands of YouTube views. This role involved creating efficient workflows with Adobe tools and ensuring consistent branding across multiple video episodes. ([HP](#), [Youtube](#))

HOKUDAI TECH GARAGE: VR Project

Mar – Apr 2022

(C#, Unity, Oculus Quest 2)

I developed a VR application that simulates unique animal sensory experiences (e.g., snake pit organs) for the Oculus

Quest 2. By conducting user interviews at a local zoo, I refined the system to better address educational needs. This project showcased my ability to deliver a complete user-centered development process, from hardware and software setup to iterative design based on real-world feedback.

Gifu City Mayor PR Video

2021

(*Adobe Premiere, After Effects*)

I produced a promotional video that highlighted local appeal and the Mayor's campaign initiatives. Handling everything from filming and editing to post-production, I iterated the final output through direct client feedback, ensuring the message aligned with the city's goals.

Personal ML for Stock Price Prediction

2020

(*Python, LightGBM, PCA, ICA*)

I built a stock price prediction system using LightGBM and incorporated feature extraction techniques such as principal component analysis (PCA) and independent component analysis (ICA). This allowed me to refine predictions in a real-world context, demonstrating my capacity to integrate multiple ML methods effectively.

FELLOWSHIPS & AWARDS

1. Hokkaido University EXEX Doctoral Fellowship (2024–2027) - \$3,000 annual + stipend
2. HOKUDAI TECH GARAGE Project Grant (2022) - \$2,000 per project

EDUCATION

Hokkaido University, Japan

April 2024 — (Expected)

Doctor of Computer Science Advisor: Prof. Daisuke Sakamoto

Hokkaido University, Japan

April 2022 – March 2024

Master of Computer Science Advisor: Prof. Daisuke Sakamoto

Hokkaido University, Japan

April 2020 – March 2022

Bachelor of Computer Science Advisor: Prof. Daisuke Sakamoto

National Institute of Technology (KOSEN), Gifu College, Japan

April 2015 – March 2020

Associate Degree of Engineering Advisor: Prof. Toshinori Deguchi

PUBLICATIONS

Peer-reviewed International Conferences

- Yuki Abe*, **Kan Kusakabe***, Myunguen Choi*, Daisuke Sakamoto, Tetsuo Ono. (*Joint first authors).
Understanding Usability of VR Pointing Methods with a Handheld-style HMD for Onsite Exhibitions.
ACM CHI 2025, April 2025. Honorable Mention Award (Acceptance rate 5%).

Peer-reviewed Domestic Journals

- **Kan Kusakabe**, Daisuke Sakamoto, Tetsuo Ono.
RingSense: Exploring User-defined Gestures for Phone Ring Holders.
情報処理学会論文誌 (IPJS), Vol. 66 (2), February 2025.
– **Hardware & ML Details:** Developed a smartphone ring accessory embedding a microcontroller and pressure sensors. Implemented a CNN-based gesture recognition system. Conducted user-defined gesture design (UDG) studies for intuitive interactions.

Peer-reviewed Domestic Conferences

- **Kan Kusakabe**, Daisuke Sakamoto, Tetsuo Ono.
スマートフォン背面のジェスチャ入力を実現するスマホリング型デバイスの設計と実装.
第30回インタラクティブシステムとソフトウェアに関するワークショップ (WISS 2022), December 2022.
(Long-press, acceptance rate: 12.5%).

Other Publications (Non-peer-reviewed)

- **Kan Kusakabe***, Yuki Abe*, Daisuke Sakamoto, Tetsuo Ono. (*Joint first author).
Game-2-X: 種類が異なるゲームプレイ間を繋ぐシステムの提案.
第 31 回インタラクティブシステムとソフトウェアに関するワークショップ (WISS 2023) , デモ発表, December 2023.
– **Concept:** Leveraged a reinforcement learning approach to “ understand ” game states and generate a cross-game scoring mechanism, allowing collaborative or competitive play across different game genres by vectorizing game states.
- **Kan Kusakabe**, Myungguen Choi, Daisuke Sakamoto, Tetsuo Ono.
無段階調整インタフェースのためのハンドジェスチャによる操作手法の探索的研究.
情報処理学会 研究報告 HCI, 2022-HCI-197(25), 1-8, March 2022.